ADITHYA VIMUKTHI

Robotics and AI Engineer | Mechatronics | Automation | Computer Vision

Sunnybank QLD 4109, Australia | +61 489 201 661 | adithyavimukthi23@gmail.com

LinkedIn: Linkedin – Adithya Vimukthi **Portfolio**: Adithya Portfolio

PROFESSIONAL SUMMARY

Mechatronics engineer with industry experience in industrial automation, machine vision, embedded systems, mechanical prototyping and system integration. Completing a Master of Advanced Robotics and Artificial Intelligence at Queensland University of Technology (QUT), with practical project experience in robot control, computer vision, force sensing and visual SLAM research. Skilled in Python, C++, MATLAB, OpenCV, SolidWorks and microcontroller-based development, with a hands-on approach to testing, troubleshooting and improving engineering systems.

TECHNICAL SKILLS

- **Programming**: Python, C++, MATLAB, embedded C/C++
- **Robotics and control**: Robot kinematics, inverse kinematics, motion control, trajectory planning, teleoperation, servo control, system modelling
- **Computer vision and AI**: OpenCV, HALCON, MediaPipe, image processing, machine vision, defect detection, machine learning, neural networks
- **Embedded systems and electronics**: Raspberry Pi, Arduino, ESP32, STM32, sensors, load cells, PWM control, circuit prototyping, soldering, PCB basics
- **Mechanical design and prototyping**: SolidWorks, CAD, mechanism design, 3D printing, CNC laser cutting, fabrication and assembly
- **Automation and integration**: HMI development, PLC-based automation, industrial automation, IoT monitoring, debugging, testing and performance optimisation
- **Research and tools**: ORB-SLAM3, visual SLAM, polarization imaging, Stokes parameters, CoppeliaSim, Git, Linux

PROFESSIONAL EXPERIENCE

Engineering Trainee | Twinery, Innovation by MAS
Mar 2024 – Nov 2024

- Developed and tested industrial automation solutions, including control systems and an HMI for a four mode overlock machine.
- Designed and prototyped electronic circuits and mechanical components; performed debugging, system integration and performance optimisation.
- Created mechanical CAD models in SolidWorks and supported fabrication, assembly and alignment using CNC laser cutting and workshop processes.
- Contributed to IoT-based automation for remote monitoring and control.
- Redesigned the sticker-pasting mechanism to handle stickers in multiple sizes and formats.

Engineering Trainee | Ansell Lanka (Pvt) Ltd
Feb 2023 – Aug 2023

- Developed two image-processing algorithms to detect shallow-depth and deep-depth product defects; trained machine-learning models, tested the inspection pipeline in HALCON, and integrated it into an automated optical inspection (AOI) system.
- Developed a PLC ladder program for a semi-automated irradiator material handling system. The program was not tested during the internship because the required components had not yet arrived.
- Participated in lean-process, CNC operation, engineering services and industrial safety activities, including solar installation inspections.

SELECTED ENGINEERING PROJECTS

Epidural Ninja — Force-Based Robotic Needle Insertion Research | QUT

- Developed a Python control stack for a 7-degree-of-freedom OpenArm, including CAN-FD motor interfacing, inverse kinematics with tool-axis constraints, straight-line needle motion and data logging.
- Evaluated force-sensing approaches for bone-contact and loss-of-resistance detection; a load cell in the needle load path distinguished printed vertebra from silicone in 26/26 insertions.
- A 100 g plunger load cell distinguished occluded and epidural-space test conditions in 25/25 trials in the tested phantom setup.

Polarization Image Fusion for Visual SLAM Under Extreme Lighting | QUT

- Developing an evaluation pipeline to test whether polarization cues improve ORB-SLAM3 robustness under saturation and harsh illumination
- Working with analyzer images at 0°, 45°, 90° and 135° to derive Stokes parameters, degree of linear polarization (DoLP) and angle of linear polarization (AoLP).
- Comparing intensity-only and image-fusion inputs using trajectory coverage, absolute trajectory error, relative pose error and tracking failures.

Real-Time Upper-Body Motion Tracking for Robot Teleoperation | Undergraduate Research

- Built a Python, OpenCV and MediaPipe-based vision system to estimate upper-body pose and arm joint angles for real-time robotic teleoperation.
- Integrated a Raspberry Pi 4-based manipulator control setup with a camera, PWM controller, servos and gripper to reproduce human arm movements.
- Co-authored a conference paper presented at the 2023 Moratuwa Engineering Research Conference (MERCon).

Industrial Automation and Mechanical Design Projects | Twinery

- Designed and integrated an HMI with four operating modes for an overlock-machine automation system.
- Contributed to mechanical design and prototyping for wire bending and multi-size sticker-pasting equipment.

EDUCATION

Master of Advanced Robotics and Artificial Intelligence | Queensland University of Technology (QUT), Brisbane

Feb 2025 – Present (Expected December 2026)

- Course GPA: 5.542/7.0 as at 25 October 2026

Bachelor of Science Honours in Engineering in Mechatronics Engineering | Sri Lanka Technological Campus (SLTC), Sri Lanka

Sep 2019 – Nov 2023

- Second Class (Upper Division) | GPA: 3.45/4.00

PUBLICATION

N. A. S. N. Nandasena, W. A. A. Vimukthi, H. M. K. K. M. B. Herath, R. Wijesinghe and S. L. P. Yasakethu, "Real-Time Upper Body Motion Tracking Using Computer Vision for Improved Human-Robot Interaction and Teleoperation," 2023 Moratuwa Engineering Research Conference (MERCon), Moratuwa, Sri Lanka, 2023.
DOI: 10.1109/MERCon60487.2023.10355479

CERTIFICATIONS AND ADDITIONAL TRAINING

- IELTS Academic — Overall Band 6.5 (July 2024)
- Modern Robotics, Course 2: Robot Kinematics — Coursera (April 2022)
- Modern Robotics, Course 1: Foundations of Robot Motion — Coursera (March 2022)
- Neural Networks and Deep Learning — Coursera (October 2021)
- Introduction to Hydraulics — Nikini Automation (2020)